

■ Table

`Table[expr, {imax}]` generates a list of *imax* copies of *expr*.

`Table[expr, {i, imax}]` generates a list of the values of *expr* when *i* runs from 1 to *imax*.

`Table[expr, {i, imin, imax}]` starts with *i* = *imin*. `Table[expr, {i, imin, imax, di}]` uses steps *di*.

`Table[expr, {i, imin, imax}, {j, jmin, jmax}, ...]` gives a nested list. The list associated with *i* is outermost.

Table evaluates its arguments in a non-standard way (see page ??). ■ Example:

`Table[f[i], {i, 4}]` $\longrightarrow$ {f[1], f[2], f[3], f[4]}. ■ Table uses the standard *Mathematica* iteration specification.

■ Example: `Table[i-j, {i, 2}, {j, 2}]` $\longrightarrow$ {{0, -1}, {1, 0}}. ■ You can use Table to build up vectors, matrices and tensors. ■ See page 90. ■ See also: `Range`, `DiagonalMatrix`, `IdentityMatrix`, `Array`, `Do`, `Sum`, `Product`.